

1 Methods

MuJoCo-Orbit couples an Earth-orbit propagator to MuJoCo’s articulated rigid-body dynamics. The chief spacecraft state $(\mathbf{R}, \mathbf{V})$ is stored in ECI coordinates with km and km/s units. MuJoCo itself is run in SI units in a chief-centered world frame whose axes remain parallel to ECI. Thus, for body i with MuJoCo world position and velocity $(\mathbf{r}_i, \mathbf{v}_i)$,

$$\mathbf{R}_i = \mathbf{R} + 10^{-3}\mathbf{r}_i, \quad \mathbf{V}_i = \mathbf{V} + 10^{-3}\mathbf{v}_i, \quad (1)$$

are the body’s absolute ECI position and velocity. Because the MuJoCo world frame translates with the chief but does not rotate, the only inertial correction required in the multibody dynamics is the translational pseudo-force associated with the chief acceleration.

1.1 Reference Orbit and Gravity

The chief is propagated as a point mass under Earth’s gravity and a swarm-mean non-gravity acceleration,

$$\dot{\mathbf{R}} = \mathbf{V}, \quad \dot{\mathbf{V}} = \mathbf{g}(\mathbf{R}) + \mathbf{a}_{\text{fb}}. \quad (2)$$

The gravity model is two-body gravity with an optional J_2 correction,

$$\mathbf{g}(\mathbf{R}) = -\frac{\mu}{r^3}\mathbf{R} + \frac{3\mu J_2 R_\oplus^2}{2r^5} \begin{bmatrix} R_x(5R_z^2/r^2 - 1) \\ R_y(5R_z^2/r^2 - 1) \\ R_z(5R_z^2/r^2 - 3) \end{bmatrix}, \quad r = \|\mathbf{R}\|. \quad (3)$$

where μ is Earth’s gravitational parameter, $R_\oplus$ is the equatorial radius, and J_2 is the second zonal coefficient. When J_2 is disabled, the second term in (3) is omitted. The feedback acceleration is not a physical force applied to every body; it only chooses a convenient moving origin. With total modeled mass $m_{\text{tot}} = \sum_i m_i$,

$$\tilde{\mathbf{a}}_{\text{fb}} = \frac{1}{m_{\text{tot}}} \sum_i \mathbf{F}_i^{\text{ng,trans}}, \quad \mathbf{a}_{\text{fb}} = 10^{-3}\tilde{\mathbf{a}}_{\text{fb}}, \quad (4)$$

where $\mathbf{F}_i^{\text{ng,trans}}$ contains drag, solar radiation pressure, and thruster forces in SI units.

1.2 Body Wrenches

For each MuJoCo body, MuJoCo-Orbit computes an external force in the chief-centered world frame and hands it to MuJoCo as a passive wrench. In SI units, the translational component is

$$\mathbf{F}_i = m_i 10^3 [\mathbf{g}(\mathbf{R}_i) - \mathbf{g}(\mathbf{R})] - m_i \tilde{\mathbf{a}}_{\text{fb}} + \mathbf{F}_i^{\text{ng}}. \quad (5)$$

The first term is differential gravity, the second compensates for the non-gravitational acceleration of the chosen origin, and $\mathbf{F}_i^{\text{ng}}$ contains body-level environmental and actuator forces. No Coriolis, centrifugal, or Euler-frame terms appear because the world axes are inertially aligned.

Surface drag and solar radiation pressure are evaluated at each surface center of pressure. For a surface normal $\hat{\mathbf{n}}_k^w$, area A_k , drag coefficient C_d^k , reflectivity coefficient C_r^k , and relative atmospheric velocity $\mathbf{v}_{ik}^{\text{rel}}$ in m/s,

$$\mathbf{F}_{ik}^{\text{drag}} = -\frac{1}{2}\rho(\mathbf{R})C_d^k A_k \max(0, \hat{\mathbf{n}}_k^w \cdot \hat{\mathbf{v}}_{ik}^{\text{rel}}) \|\mathbf{v}_{ik}^{\text{rel}}\| \mathbf{v}_{ik}^{\text{rel}}, \quad (6)$$

$$\mathbf{F}_{ik}^{\text{SRP}} = -\eta P_\odot C_r^k A_k \max(0, \hat{\mathbf{n}}_k^w \cdot \hat{\mathbf{s}}) \hat{\mathbf{s}}. \quad (7)$$

Here ρ is atmospheric density, η is the eclipse factor, $P_\odot = 4.56 \times 10^{-6} \text{ N/m}^2$, and $\hat{s}$ is the Sun direction. The surface moment arm supplies the corresponding torque about the body COM. Atmospheric density, magnetic field, Sun direction, and eclipse are cached at the chief state; body position, velocity, attitude, and surface moment arms are evaluated per body.

The rotational wrench includes finite-body gravity-gradient torque, magnetic torques, actuator gyroscopic torques, and force-offset torques:

$$\boldsymbol{\tau}_i = \boldsymbol{\tau}_i^{gg} + \boldsymbol{\tau}_i^{\text{surf}} + \boldsymbol{\tau}_i^{\text{mag}} + \boldsymbol{\tau}_i^{\text{act}} + \boldsymbol{\tau}_i^{\text{thr}}. \quad (8)$$

The gravity-gradient term is

$$\boldsymbol{\tau}_i^{gg} = \frac{3\mu}{\|\mathbf{R}_i\|^3} \hat{\mathbf{R}}_i \times \left(J_i^w \hat{\mathbf{R}}_i \right), \quad \hat{\mathbf{R}}_i = \mathbf{R}_i / \|\mathbf{R}_i\|, \quad (9)$$

where J_i^w is the inertia tensor in world axes. Residual dipoles and magnetorquers use $\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}$ in the body frame and rotate the result to world coordinates. Reaction wheels and single-gimbal CMGs apply the standard gyroscopic reaction torques from their stored momentum states, while thrusters apply both a directed force and its moment about the body COM.

1.3 Multibody Coupling

MuJoCo advances the generalized multibody state $(q, \dot{q})$, including free-joint roots, articulated joints, equality constraints, and contacts. With MuJoCo's built-in uniform gravity disabled, the constrained dynamics can be written

$$M(q)\ddot{q} + \mathbf{c}(q, \dot{q}) = \boldsymbol{\tau}^{\text{passive}} + \boldsymbol{\tau}^{\text{ctrl}} + J_c^\top \boldsymbol{\lambda}, \quad (10)$$

where $\boldsymbol{\lambda}$ denotes the contact and constraint impulses solved by MuJoCo. MuJoCo-Orbit injects the orbit/environment effects through `qfrc_passive` using `mj_applyFT`. Equivalently,

$$\boldsymbol{\tau}^{\text{passive}} = \sum_i \left(J_i^\top \mathbf{F}_i + J_i^{R\top} \boldsymbol{\tau}_i \right), \quad (11)$$

with J_i and J_i^R the translational and rotational body Jacobians. This projection makes the same per-body wrench formulation valid for free bodies, articulated spacecraft, and contact-coupled systems. Contact impulses are internal to the MuJoCo solve and are not included in the chief feedback acceleration in (4).

1.4 Step Procedure

The coupling is implemented as a MuJoCo plugin attached during model compilation. A call to `mjo_step` executes four stages:

1. update environment caches from the current chief state;
2. assemble body forces and torques from (5)–(8) into `qfrc_passive`;
3. let MuJoCo solve and integrate the constrained multibody dynamics;
4. advance stored actuator states and propagate the chief state using (2), then refresh post-step caches.

For comparisons with external simulators, MuJoCo states are converted back to absolute ECI coordinates using (1); derived LVLH helper transforms are used only for initialization, analysis, and visualization.